

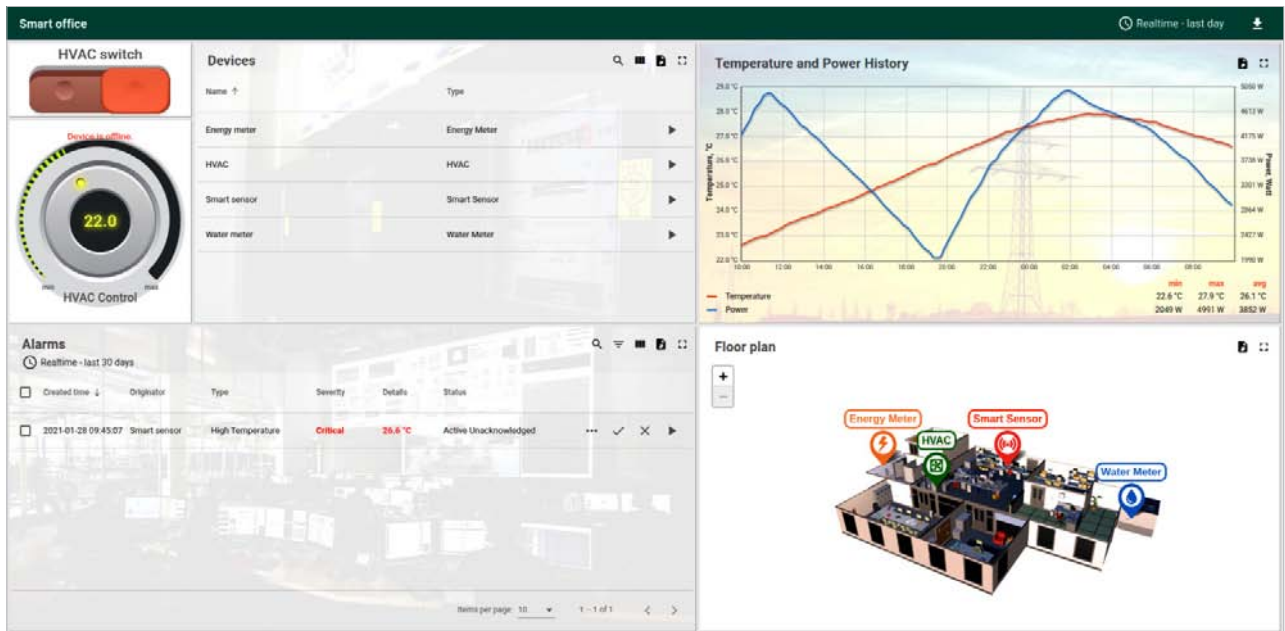


Figure 3: Smart office dashboard

Open source intelligent office solutions

There are many open source platforms available that can be used to build intelligent office solutions. Mainflux, ThingsBoard, DeviceHive, Thinger, Zetta, OpenRemote, Kaa, Fiware, ThingsConnect, WS02Rap are popular open source IoT platforms.

ThingsBoard: This platform dramatically reduces the effort needed to create smart office solutions. It is widely used for:

- Smart energy management.
- Access logging and control
- Indoor tracking of employees
- Occupancy of meeting and wash rooms
- Indoor air quality and HVAC monitoring

By leveraging this IoT platform, we can save up to 90 per cent of the development time needed to build smart office solutions. The benefits are:

- Reliable and fault-tolerant data collection from IoT devices and sensors
- Powerful rule engine to process collected data for valuable insights like setting of alarms

- Advanced and flexible visualisation of real-time and historical data
- Customisable end user dashboards to share the monitoring results
- On-premises and cloud deployment options
- Remote control and OTA (over the air) updates for IoT devices
- Customisable mobile applications with minimum coding efforts

ThingsBoard provides production-ready server infrastructure to connect IoT devices, as well as store, analyse and share the collected data.

The ThingsBoard smart office template represents a basic workspace monitoring and management solution. With this layout (Figure 2), we get an interactive dashboard with the ability to control the HVAC system, and have a high-level report of the key

metrics necessary for effective and proactive office management. As part of this solution, there are smart office dashboards that display data from multiple devices. These dashboards may be used to:

- Observe office sensors and their location
- Browse indoor temperature and power consumption history
- Monitor temperature alarms
- Control HVAC (requires connected device)
- Observe specific details for each sensor

IoT devices also accept commands from the dashboard. For example, HVAC sensors accept commands to enable/disable air conditioning as well as set target temperature. The commands are sent using the platform API. **END** 🐧

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